Course Code	**:**	**MCS-224**
Course Title	**:**	**Artificial Intelligence and Machine Learning**
Assignment Number	**:**	**MCAOL(III)/224/Assign/2023**
Maximum Marks	**:**	**100**
Weightage	**:**	**30%**
Last date of Submission	**:**	**30th April, 2023 (for January session)** **31st October, 2023 (for July session)**

This assignment has sixteen questions of 5 Marks each, answer all questions. Rest 20 marksare for viva voce. You may use illustrations and diagrams to enhance the explanations. Please go through the guidelines regarding assignments given in the Programme Guide for the format of presentation.

Q1. Differentiate among Descriptive, Predictive and Prescriptive analytics performed under Machine Learning.

Q2. What are Intelligent agents in AI? Briefly discuss the properties of Agents.

Q3. Find the minimum cost path for the 8-puzzle problem, where the start and goal state are given as follows:

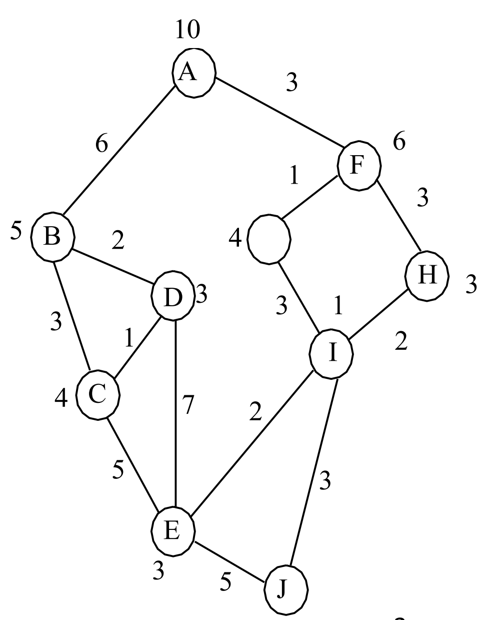
1	2	3
4	8	-
7	6	5

Start State

1	4	7
2	8	6
3	-	5

Goal State

Q4. Consider the following graph. The numbers written on edges represents the distance between the nodes and the numbers written on nodes represents the heuristic value. Find the most cost-effective path to reach from Noda A to node J using A* Algorithm.



Q5. Discuss the transforming an FOPL Formula into Prenex Normal Form with suitable example. Also, discuss Skolomization with a suitable example.

Q6. Explain Forward Chaining Systems and Backward Chaining Systems with a suitable example for each.

Q7. Draw a semantic network for the following English statement:
Shyam struck Neha and Neha's father struck Shyam.

Q8: Write short notes on following
a) Reinforcement Learning b) Ensemble method

Q9. Explain the phases of machine learning cycle

Q10. Explain working of Back Propagation and Convolution Neural Network

Q11. What is pattern search? Discuss the Apriori Algorithm for pattern search.

Q12. Explain Naïve Bayes Classification Algorithm with a suitable example.

Q13. Explain K-Nearest Neighbors classification Algorithm with a suitable example.

Q14. For the given points of two classes red and blue:
Blue: { (1, 2), (2,1), (1,-2), (2,-2)}
Red : { (4,-1), (4,1), (5,-1), (6,1)}

Plot a graph for the red and blue categories. Find the support vectors and optimal separating line.

Q15. Explain PCA with a suitable example.

Q16. Explain A priory Algorithm with a suitable example